

**Real feature
geometry**

X , groups g (wheat,
rice, microbiome, drug)

**Inject synthetic
truth**

Type E pool L ·
Type A shortcut η

Stored truth file

expected Type A /
Type E status

**Detector
evaluation**

FPR · power · LOD
vs known truth

Original biological labels are not used as truth — only feature geometry and group structure are reused.